SPECTRA

Infrared Image Super-Resolution & Colorization Pipeline

Master Reference Document — PS10, Bharatiya Antariksh Hackathon 2026

Problem Statement	PS10 — Infrared Image Colorization and Enhancement for Improved Object Interpretation
Project Codename	SPECTRA
Team	Team STARCY — Saraswati College of Engineering, Navi Mumbai
Idea Submission Deadline	1st July 2026
Grand Finale	6th – 7th August 2026
Document Purpose	Single source of truth covering scope, architecture, datasets, models, UI, evaluation criteria, and build checklist — nothing should be missed.

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1. What Changed After the Explainer Session

Our original plan (built before the 16th June Problem Statement Explainer Session) was based on a generic understanding of “IR image colorization.” After ISRO's official session, the scope is more specific and technically demanding. This section is the most important — read this first.

1.1 Original Plan vs Official ISRO Requirement

Aspect	Our Original Plan	Official ISRO Requirement
Core task	Single-stage colorization only	Two-stage: Super-Resolution (200m→100m) + Colorization (TIR→RGB)
Dataset	FLIR ADAS Thermal Dataset (automotive)	Landsat 9 (USGS) — Bands B2, B3, B4, B10
Domain	Street-level dashcam thermal imagery	Satellite earth-observation imagery
Output count	One output (colorized image)	Two outputs: high-res TIR @100m AND colorized RGB @100m
Evaluation	Visual judgement only	PSNR, SSIM, FID + qualitative hallucination check + inference time
Bonus scope	Not considered	Physics-informed modeling (blackbody/emissivity-based constraints)
Reference code	None	Official GitHub repo (currently empty, awaiting ISRO's script)

1.2 Verdict: Does the Path Change?

- **Stays the same:** Generative deep learning approach (GAN-based), FastAPI + React architecture, before/after comparison as the demo centerpiece, SPECTRA as project name and brand identity.
- **Changes:** Must become a two-model pipeline (Super-Resolution model + Colorization model chained together), dataset switches entirely to Landsat 9 satellite bands, evaluation metrics must be visible in the product itself (not just claimed in the report), and an end-to-end reproducible Python pipeline (download → preprocess → infer → evaluate) is explicitly expected.

2. Official Problem Definition (As Given by ISRO/SAC)

2.1 Official Title

“PS10: Infrared Image Colorization and Enhancement for Improved Object Interpretation” — presented by ISRO Team: Ashutosh Gupta, Hemant Lalwani, Jugal Patel (SAC — Space Applications Centre), 16th June 2026.

2.2 Expected Input / Output

Item	Specification
Input	Single-channel TIR (Thermal Infrared) image at 200m resolution
Output 1	High-resolution TIR image at 100m resolution (super-resolved, still single-channel grayscale)
Output 2	Colorized RGB image at 100m resolution (final human-interpretable output)
Deliverable	An end-to-end trained Deep Learning pipeline, Python-based, fully reproducible from raw data to final output

2.3 Official 4-Step Workflow (from ISRO slides)

1. **Step 1 — Input:** TIR satellite imagery at 200m resolution
2. **Step 2 — Super Resolve:** Super-resolution model upscales 200m → 100m (still grayscale TIR)
3. **Step 3 — Colorize:** Colorization model converts the 100m grayscale TIR into 100m RGB
4. **Step 4 — Compare:** Output is compared against the original real TIR at 100m and real RGB reference imagery

2.4 Official Dataset Source

- **Source:** Landsat 9 (USGS), processed TIR1–RGB pairs using ISRO's GitHub instructions.
- **Bands required:** B2 (Blue), B3 (Green), B4 (Red) — merged into RGB; B10 (TIRS-1, Thermal Infrared) — native 100m resolution sensor.
- **Reference repo:** github.com/jugal-sac/IR-colorization-BAH2026 — currently contains only a README, no scripts pushed yet (status as of 17th June 2026). Must be monitored (Watch button) for updates.
- **Approved tools:** GDAL, Rasterio, tifffile, OpenCV — explicitly named in ISRO's tech stack slide.
- **Approved model families:** GANs, Diffusion Models, or other State-of-the-Art Image-to-Image models — explicitly named.

2.5 Official Dataset Preparation Workflow

- Download Landsat 9 Bands: B2, B3, B4, B10 for the chosen region.
- Merge B2 + B3 + B4 into a single real RGB image.
- Downscale that RGB image by 3.3× to simulate 100m resolution (matches TIR's native scale).
- Take B10 (TIR) and downscale by 3.3× → simulates 100m ground truth TIR.
- Take B10 (TIR) and separately downscale by 6.7× → simulates 200m, this becomes the LOW-RES INPUT.
- Create image patches (small tiles) from both the 100m and 200m versions — not full satellite scenes.
- Train Super-Resolution model: 200m TIR patch → 100m TIR patch.
- Train Colorization model: 100m TIR patch → 100m RGB patch.

2.6 Official Evaluation Parameters — This Is How You Win or Lose

Category	Metric	What It Means For Us
Image Quality	**PSNR**	Peak Signal-to-Noise Ratio — pixel-level accuracy of our output vs ground truth. Higher is better.
Image Quality	**SSIM**	Structural Similarity Index — measures structural/perceptual closeness, not just pixel values.
Image Quality	**FID**	Fréchet Inception Distance — measures how realistic our generated image distribution looks overall (key GAN quality metric).
Qualitative	**Visual Inspection**	Judges manually check for 'hallucinations' — i.e. the model inventing objects/structures (fake roads, buildings) that do not exist in reality. Must be avoided.
Preferred	**Inference Time / Tile**	Lower is better. A lightweight, fast model is explicitly preferred over a slow, heavy one.
Bonus	**Physics-Informed Modeling**	Extra credit for incorporating real thermal physics (blackbody radiation, emissivity) into the model rather than a pure black-box approach.

3. What Stays From Our Original SPECTRA Plan

Not everything changes. These elements from our original planning remain valid and should be carried forward as-is.

- **Project name & brand:** "SPECTRA" stays as the official project name. Logo direction (prism splitting grayscale beam into color spectrum, deep space black background, electric blue accent) stays.
- **Core ML family:** GAN-based image-to-image translation (pix2pix-style Generator + PatchGAN Discriminator) remains the right approach for the colorization stage specifically.
- **Loss function logic:** Adversarial Loss + L1 Loss + SSIM/PSNR evaluation — this combination was already part of our original plan and is now explicitly validated by ISRO's own evaluation parameters slide.
- **Backend framework:** FastAPI (Python) for serving the model remains correct and unchanged.
- **Frontend framework:** React + TypeScript + Vite remains correct and unchanged.
- **UI centerpiece concept:** Before/after comparison slider remains the right idea for the demo — it now simply needs to expand from a 2-way comparison to a 4-way comparison (see Section 6).
- **Deployment plan:** Vercel (frontend) + Railway/Render (backend) remains valid and unchanged.

4. New Requirements to Implement Now

These are the additions/changes mandated directly by the explainer session that were not part of the original plan.

4.1 Two-Model Pipeline Instead of One

- **Model A — Super-Resolution Model (NEW):** Input: 200m TIR tile. Output: 100m TIR tile. This model did not exist in our original plan at all.
- **Model B — Colorization Model (carried forward, repositioned):** Input: 100m TIR tile (output of Model A, not raw input). Output: 100m RGB tile. This was our entire original plan; now it is Stage 2 of a 2-stage pipeline.
- **Pipeline chaining:** Model A's output must feed directly into Model B as input — they cannot be trained or evaluated in total isolation from each other, since the final deliverable is the chained result.

4.2 Dataset Migration

- **Drop entirely:** FLIR ADAS Thermal Dataset (automotive dashcam domain — wrong sensor characteristics, wrong resolution scale, wrong viewing angle for satellite imagery).
- **Adopt:** Landsat 9 via USGS EarthExplorer (earthexplorer.usgs.gov) and/or Google Earth Engine (earthengine.google.com, dataset ID: LANDSAT/LC09/C02/T1_L2) — both are legitimate access points to the same official USGS data.
- **Pending clarification:** Official preprocessing script from github.com/jugal-sac/IR-colorization-BAH2026 not yet published. Email sent to Hack2skill support; also posted on Discord. Action: check repo and Discord daily for updates.
- **Risk mitigation:** Build our own data pipeline now using EarthExplorer/Earth Engine, but structure the code so the input is simply ‘a folder of GeoTIFF band files’ — this way it slots into the official script later regardless of which exact tool ISRO's instructions reference.

4.3 Visible Evaluation Metrics (Mandatory, Not Optional)

- **In the report/slide:** Static 4-panel comparison image with PSNR/SSIM/FID values labeled under each output.
- **In the live product:** Real-time computed PSNR, SSIM, FID displayed in the UI for every processed image — this was not part of the original plan and must be added to the backend response and frontend display.
- **Inference timer:** Must display per-tile inference time in the UI, since ISRO explicitly listed this as a preferred criterion.

4.4 Physics-Informed Modeling (Bonus — Differentiator)

- **Concept:** Most competing teams will likely submit a pure black-box GAN/diffusion approach. Incorporating real thermal physics gives a competitive edge.
- **Practical approach:** Add a physics-consistency loss term that penalizes RGB-temperature combinations that are physically implausible given blackbody radiation/emissivity principles for the materials typically present in the scene (water, vegetation, concrete, metal roofing).
- **Scope note:** This is a bonus — implement only after Models A and B are working end-to-end. Do not let this block the core deliverable.

4.5 End-to-End Reproducible Python Pipeline

- **Requirement:** A single, runnable Python pipeline: raw Landsat band download → preprocessing (merge, downscale, patch) → Model A inference → Model B inference → metric evaluation → final output. Not just a Jupyter notebook with disconnected cells.
- **Why this matters:** ISRO's own slide literally says 'An end-to-end trained Deep Learning pipeline' as the expected output — this is a direct evaluation criterion, not a nice-to-have.

5. Complete Tech Stack

Layer	Technology
Data Source	Landsat 9 (USGS) — Bands B2, B3, B4, B10, via EarthExplorer and/or Google Earth Engine API
Data Processing Libraries	GDAL, Rasterio, tifffile, OpenCV, NumPy (all explicitly approved by ISRO)
Model A — Super-Resolution	PyTorch — SRGAN / ESPCN / EDSR-style architecture
Model B — Colorization	PyTorch — pix2pix GAN (U-Net Generator + PatchGAN Discriminator)
Training Environment	Google Colab (free T4 GPU) for initial training; local/Kaggle as backup
Evaluation Metrics	scikit-image (PSNR, SSIM), pytorch-fid or clean-fid (FID)
Backend	Python, FastAPI, Uvicorn
Frontend	React, TypeScript, Vite
Deployment	Vercel (frontend), Railway or Render (backend)
Branding	Project name: SPECTRA. Visual identity: prism/spectrum motif, deep space black + electric blue palette

6. UI / Frontend — What Must Be Built

6.1 Upload Zone

- Drag-and-drop or click-to-upload for the raw 200m TIR input image/tile.
- Client-side file type and size validation before sending to backend.
- Immediate preview of the uploaded raw input.

6.2 Processing View

- Loading state with progress indication while Model A (super-resolution) and Model B (colorization) run sequentially.
- Live inference timer counting elapsed processing time per tile.

6.3 Four-Way Comparison — The Core Demo Component

This is the single most important UI component, directly mirroring ISRO's own 'Step 4: Compare' slide. It must show all four images, not just two:

- Panel 1 — Low-resolution input TIR (200m, grayscale)
- Panel 2 — Super-resolved TIR (100m, grayscale, our Model A output)
- Panel 3 — Colorized RGB (100m, our Model B output — the final prediction)
- Panel 4 — Real RGB reference (100m, ground truth from Landsat 9 — for comparison only, not part of our pipeline's output)
- **Interaction style:** Slider or toggle to move between panels, or a 4-up grid view depending on screen space; should work both as a static exportable image (for the report/slide) and an interactive live version (for the deployed website).

6.4 Metrics & Results Panel

- PSNR score, displayed numerically with a clear label.
- SSIM score, displayed numerically with a clear label.
- FID score, displayed numerically with a clear label (computed at batch/dataset level since FID is not a single-image metric — must be clearly explained in UI as a dataset-level score, not per-image).
- Inference time per tile, in milliseconds or seconds.
- Download button for both outputs (super-resolved TIR and colorized RGB).
- 'Try another image' reset action.

6.5 Visual Design Direction

- Dark background to make colorized outputs visually pop against the interface.
- Electric blue accent color for interactive elements, consistent with SPECTRA brand identity.
- Clean, modern, scientific feel — avoid cluttered dashboards; prioritize the comparison view as the visual hero of the page.

7. Backend — What Must Be Built

7.1 API Endpoints

Endpoint	Function
POST /process	Accepts a 200m TIR tile, runs Model A then Model B sequentially, returns all four images (input, super-resolved, colorized, and reference RGB if available) plus PSNR/SSIM/FID/inference-time metrics
GET /health	API status check
GET /metrics/{id}	Fetch stored evaluation metrics for a previously processed tile

7.2 Processing Pipeline Inside the Backend

5. Receive and validate uploaded TIR tile (format, single-channel check, expected resolution).
6. Preprocess: normalize pixel values, resize/pad to model's expected tile size.
7. Run Model A (super-resolution): 200m → 100m grayscale TIR.
8. Run Model B (colorization): 100m grayscale TIR → 100m RGB.
9. Postprocess: denormalize tensors back to image format (PNG).
10. If ground-truth RGB is available for that tile (test set), compute PSNR, SSIM against it; FID computed at batch level across the test set.
11. Return JSON response with base64-encoded images and all metric values plus total inference time.

8. ML Model Architecture Detail

8.1 Model A — Super-Resolution

- **Architecture options:** SRGAN, ESPCN, or EDSR — pick based on time available; ESPCN is the lightest and fastest to train for a hackathon timeline.
- **Input:** 200m single-channel TIR patch.
- **Output:** 100m single-channel TIR patch (same content, higher spatial resolution).
- **Loss:** Pixel-wise L1/L2 loss plus optional perceptual loss; evaluated with PSNR and SSIM against the real 100m TIR.

8.2 Model B — Colorization

- **Architecture:** pix2pix — U-Net Generator (encoder-decoder with skip connections) + PatchGAN Discriminator (70×70 patches, classifies real vs fake locally).
- **Input:** 100m single-channel TIR patch (output of Model A).
- **Output:** 100m RGB patch.
- **Loss:** Adversarial loss (from discriminator) + L1 loss (pixel accuracy against real RGB); evaluated with SSIM and FID.

8.3 Training Data Construction (Per Section 2.5)

- Real RGB tile (from B2+B3+B4, downscaled 3.3×) acts as ground truth for Model B.
- Real TIR tile at 100m (B10, downscaled 3.3×) acts as ground truth for Model A's output and as Model B's training input.
- Simulated 200m TIR tile (B10, downscaled 6.7×) acts as Model A's training input.
- All tiles must be small patches, not full satellite scenes, to keep training memory and time manageable.

8.4 Bonus: Physics-Informed Constraint

- Add an auxiliary loss term that uses approximate blackbody radiation/emissivity relationships to constrain plausible RGB outputs for a given thermal intensity and likely surface material (water, vegetation, urban concrete, bare soil).
- This should be implemented as an additional loss term added to Model B's existing loss function, not a separate model — keep it lightweight and additive.

9. Master Build Checklist — Nothing Should Be Missed

9.1 Data Pipeline

- Confirm final dataset source (own EarthExplorer/Earth Engine pipeline vs official ISRO script) once GitHub repo is updated or Hack2skill responds.
- Download Landsat 9 scenes for at least 2–3 diverse Indian regions (urban, rural/vegetation, water body) for variety.
- Write band-merge script (B2+B3+B4 → RGB).
- Write downscaling script (3.3× and 6.7× variants of B10).
- Write patch-extraction script (small tiles from full scenes).
- Create train/validation/test split.

9.2 Model A — Super-Resolution

- Implement architecture (SRGAN/ESPCN/EDSR).
- Train on 200m→100m TIR patch pairs.
- Evaluate with PSNR/SSIM, save best checkpoint.

9.3 Model B — Colorization

- Implement pix2pix (U-Net Generator + PatchGAN Discriminator).
- Train on 100m TIR → 100m RGB patch pairs.
- Evaluate with SSIM/FID, save best checkpoint.
- (Bonus) Add physics-informed loss term.

9.4 Backend

- Build /process endpoint chaining Model A → Model B.
- Compute and return PSNR/SSIM/FID/inference time in response.
- Test with Postman/cURL before connecting frontend.
- Deploy on Railway or Render.

9.5 Frontend

- Build upload component with validation and preview.
- Build 4-way comparison view (input, super-resolved, colorized, real RGB).

- Build metrics display panel (PSNR, SSIM, FID, inference time).
- Add download buttons for both outputs.
- Deploy on Vercel.

9.6 Submission Materials

- Static 4-panel comparison image with metric values, for the idea submission slide (due 1st July).
- Short technical write-up covering the two-stage pipeline, dataset preparation, and evaluation approach.
- Live demo link ready before the Grand Finale (6–7 August).
- SPECTRA logo and branding finalized for slide/demo polish.

9.7 Open Items to Track

- **Pending:** Official GitHub repo (github.com/jugal-sac/IR-colorization-BAH2026) is empty as of 17th June 2026 — check daily for the actual preprocessing script.
- **Pending:** Awaiting response from Hack2skill support on whether independent EarthExplorer/Earth Engine sourcing is acceptable in the interim.
- **Action:** Monitor official Discord server for announcements and other teams' clarifying questions on this same topic.